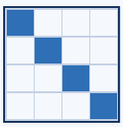
Ranks readable from structure

For these types the rank follows from the defining condition, so no reduction is needed. Grouped by whether the structure pins the rank exactly or only bounds it.

5
TYPES

FULL RANK, GUARANTEED 2



Identity | I_n

RANK	WHY	NULLITY
n	columns are the standard basis	0

The columns are independent by construction. Trivially the maximum, and the case every other row is measured against.

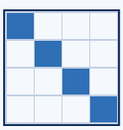


Symmetric positive definite | $\mathbf{x}^\mathsf{T} A\mathbf{x} > 0$ for $\mathbf{x} \neq \mathbf{0}$

RANK	WHY	NULLITY
n	no nonzero vector maps to zero	0

If $A\mathbf{x} = \mathbf{0}$ then $\mathbf{x}^\mathsf{T} A\mathbf{x} = 0$, which the condition forbids unless $\mathbf{x} = \mathbf{0}$. So the null space is trivial and the rank is full — an argument that never mentions rows or columns.

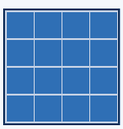
RANK FIXED BY A COUNT 2



Diagonal | $\operatorname{diag}(d_1, \ldots, d_n)$

RANK	WHY	NULLITY
count of $d_i \neq 0$	each nonzero entry is one pivot	count of $d_i = 0$

Already in echelon form, so the pivots are the nonzero diagonal entries and no reduction happens. Rank and nullity are read off the same list.

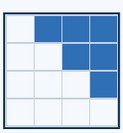


Rank-one outer product | $A = \mathbf{u}\mathbf{v}^\mathsf{T}$, both nonzero

RANK	WHY	NULLITY
1	every column is a multiple of $\mathbf{u}$	$n - 1$

Column j is $v_j\mathbf{u}$, so the column space is the single line through $\mathbf{u}$ however large the matrix is. These are the building blocks the SVD sums, and truncating that sum is low-rank approximation.

ONLY A BOUND 1



Nilpotent | $A^k = O, A \neq O$

RANK	WHY	NULLITY
$< n$ — strictly deficient	a full-rank matrix has a full-rank power	≥ 1

The condition forces deficiency but does not say by how much: a 3×3 nilpotent matrix may have rank 1 or 2. This is the row where structure narrows the answer without determining it, and the reduction still has to be done.

What the reducible cases have in common is a definition that already says how many independent directions there are. A rank-one outer product has one because every column is a multiple of $\mathbf{u}$; a diagonal matrix has as many as it has nonzero entries. Where the definition is silent about independence — as with nilpotence — only a bound survives.